

Course Syllabus: FRA503 Introduction to Machine Learning (1/2569)

Instructors:

Asst.Prof. Paisit Khanarsa, Ph.D. (Aj.Por)

Emails:

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Times:

Wednesday 13.30-16.30

Course description:

This course provides an introduction to machine learning. Topics include: (i) Introduction to machine learning (first machine learning and machine learning cycle). (ii) Supervised learning (linear regression, logistic regression, k-nn, decision tree, neural networks). (iii) Data preprocessing for machine learning (preprocessing techniques, feature selection, feature extraction) (vi) Unsupervised learning (k-mean clustering, hierarchical clustering, DBSCAN). (v) Introduction to deep learning. The course will also go through the application of machine learning by python programming so that you will also learn how to apply machine learning algorithms to datasets in various fields.

Important dates:

- Due date HW1: 30/9/2026
- Due date HW2: 4/11/2026
- Due date Project: 30/12/2026 - 12/12/2026

Grading

- Homework 60% (#HW)(1-3 people per group)
 - Code file report (.ipynb)
- Final project 40%
 - Presentation
 - Code file and report (.ipynb)

Course Calendar

FRA503 Introduction to Machine Learning				Course Calendar	
	Time	13:30-16:30 (Wednesday)			
Week	Day	Date	Instructor	Topic	HW
1	Wed	5/8/2026	Aj. Paisit	Introduction to ML, ML Cycle and First ML	
2	Wed	12/8/2026	Aj. Paisit	Break	
3	Wed	19/8/2026	Aj. Paisit	Basic Regression Problems	
4	Wed	26/8/2026	Aj. Paisit	Data Preprocessing & Feature Engineering (Part 1)	
5	Wed	2/9/2026	Aj. Paisit	Data Preprocessing & Feature Engineering (Part 2)	HW1 Assigned
6	Wed	9/9/2026	Aj. Paisit	Hands-on Coding for Basic ML Process	
7	Wed	16/9/2026	Aj. Paisit	Basic Classification Problems 01	
9	Wed	23/9/2026	Aj. Paisit	Basic Classification Problems 02 K-NN and Decision Tree 01	HW2 Assigned
10	Wed	30/9/2026	Aj. Paisit	K-NN and Decision Tree 02	Project Assigned HW1 Due
Exam1/1	Wed	12-17/10/2025		Break	
Exam1/2	Wed	19-23/10/2025		Break	
Exam1/2	Wed	28/10/2025		Break	
12	Wed	4/11/2026	Aj. Paisit	Introduction to Neural Network & Hyperparameter Tuning	HW2 Due
13	Wed	11/11/2026	Aj. Paisit	Introduction to Deep Learning	
14	Wed	18/11/2026		Break or Project Consultation	
15	Wed	25/11/2026		Break or Project Consultation	
Exam2	Wed	30/12/2026- 12/12/2026	Aj. Paisit	Final Project Presentation (Online/Onsite) Total presentation time: 30 minutes (20 minutes presentation + 10 minutes Q&A)	TBD
16	Wed	10/12/2026		Break	

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